LESSON PLAN

ENGINEERING METROLOGY (13)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	06
Total No. of Lesson Plans	9	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Hovering and forward flight, take off and climb to free air hover, vertical decent ground effect ground cushion re circulation effects, forces in balance during hover, forward flight and transition phenomenon of flap back, factors effecting maximum speed.
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Hovering and forward flight, take off and climb to free air hover, vertical decent ground effect ground cushion re circulation effects, forces in balance during hover, forward flight and transition phenomenon of flap back, factors effecting maximum speed.
5. Specific Objectives	13.1.1.1 Trainee will be able to explain the helicopter control, main rotor movements, translation flight, power, auto rotation, vibration checks and various adjustments.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of flight control stick positions and cyclic/collective inputs for hover and forward flight; Instructor supervision of theoretical application only.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What are the primary flight controls of a helicopter?
(b) Define the term 'free air hover'.
(c) List three factors affecting maximum speed.

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Forces in balance during hover • Take off and climb to free air hover • Vertical descent and ground effect	L & Ds	PPT, IB	15 min
(b)	• Ground cushion and re-circulation effects • Forward flight and transition phenomenon • Flap back and factors affecting maximum speed	L & Ds	PPT, PDF	20 min
(c)	• Helicopter control and main rotor movements • Translation flight and power management • Auto rotation and vibration checks	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain the phenomenon of flap back during forward flight.
(b) Describe the role of ground effect in vertical descent.
(c) How do vibration checks relate to main rotor movements?

12. Home Assignment and Follow Up	Prepare a brief note on power requirements, induced power, rotor, profile and parasite power to be discussed in LP 07.

Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Power requirements, induced power, rotor, profile and parasite power and total power requirements.	Refer to the master syllabus reference list	LP 07

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min